

COMPUTER NETWORKS

Introduction

- ❑ Computer networks today enable you to:
 - ❑ Send a message consisting of several pages, pictures, sounds and animations to a group of people anywhere at once and in a few minutes.
 - ❑ You connect from your home or work computer to international information banks, companies and libraries to obtain the information that interests you.
 - ❑ Holding interactive conferences and seminars for remote parties via the Internet.
 - ❑ E-Learning, E-Commerce and E-Government.
- ❑ All this would not have been possible without the integration between communication technology and computer technology and the creation of the so-called computer network.

Network

- ❑ A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications.
- ❑ The computers on a network may be linked Wire or Wireless
 - ❑ through cables, telephone lines, radio waves, satellites, or infrared light beams.

Benefits of Computer Networks

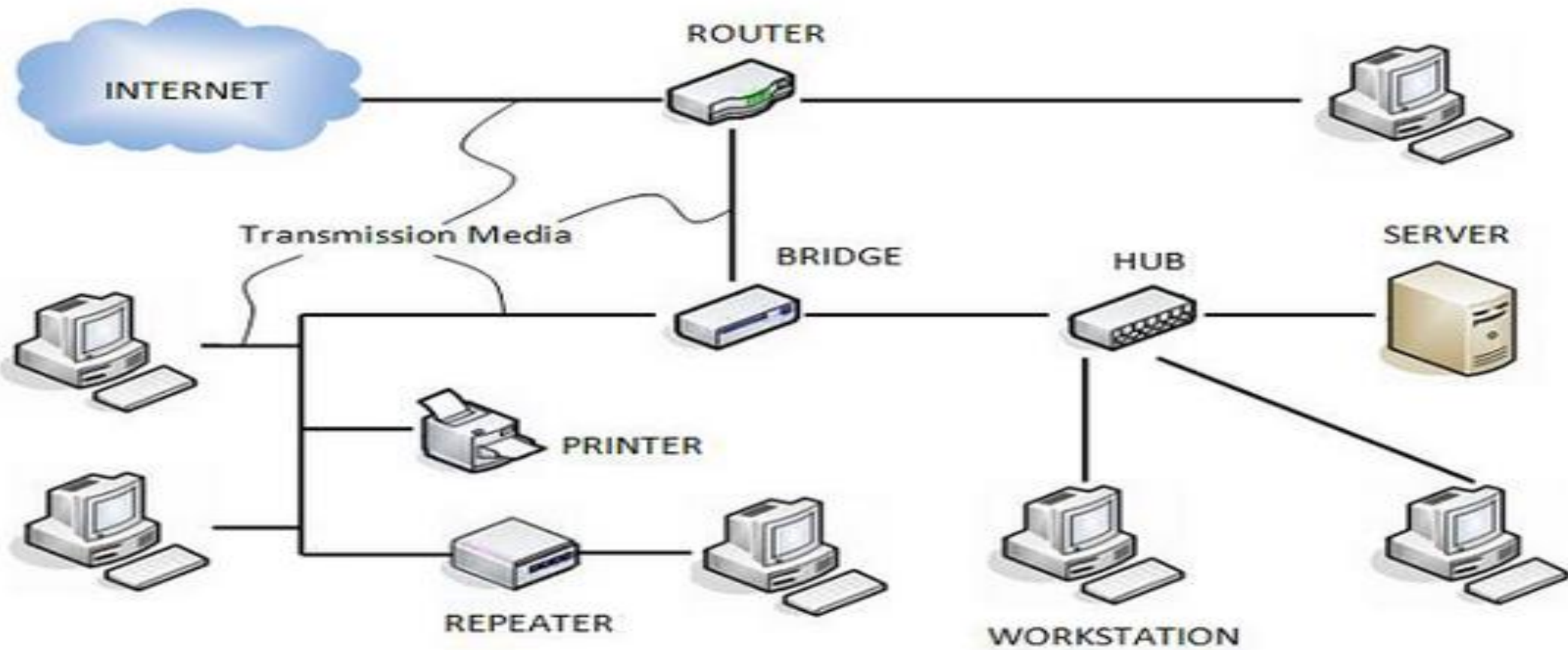
- ❑ File sharing - you can easily share data between different users, or access it remotely if you keep it on other connected devices.
- ❑ Resource sharing - using network-connected peripheral devices like printers, scanners and copiers, or sharing software between multiple users, saves money.
- ❑ information sharing and get in touch more easily.
- ❑ Business becomes more efficient
 - ❑ Ex. networked access to a common database can avoid the same data being keyed multiple times, saving time and preventing errors

Computer Network Components

□ A computer network consists of several parts, each part has its own function in the networked system, and these parts are:

- Server
- Clients
- Transmission Media
- Network Interface Card
- Peripherals
- Communication Switches
- Network Operating System





COMPUTER NETWORK COMPONENTS

Computer Network Components Cont..

- ❑ **Servers:** Servers are computers that hold shared files, programs, and the network operating system. Servers provide access to network resources to all the users of the network. It is characterized by high processing speed and large storage capacity
- ❑ **Clients (Workstations)** – Clients are computers that request and receive service from the servers to access and use the network resources.
- ❑ **Transmission media** are the channels through which data is transferred from one device to another in a network. Transmission media may be guided media like coaxial cable, fiber optic cables etc; or maybe unguided media like microwaves, infra-red waves etc.

Computer Network Components Cont..

❑ **Network Interface Card:** each computer in a network has a special expansion card called a network interface card (NIC). The NIC prepares(formats) and sends data, receives data, and controls data flow between the computer and the network.

❑ **Peripherals:** peripherals are hardware resources provided to the users of the network by servers. Resources provided include printers, scanner, fax,...

❑ **Communication Switches:** connecting devices act as middleware between networks or computers, by binding the network media together.

❑ Ex: Routers, Hubs, Switches, Repeaters

Computer Network Components Cont..

Network operating system is a program that runs on computers and servers that allows the computers to communicate over the network.

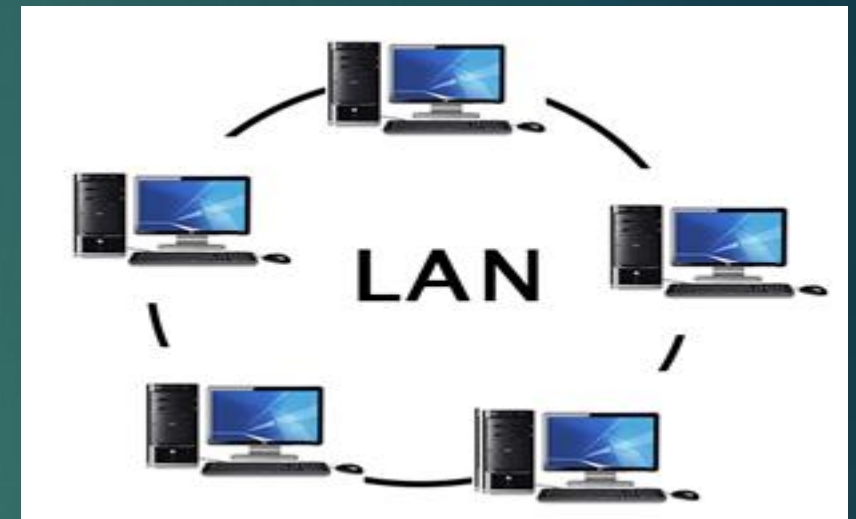
Computer Network Types

❑ A computer network can be categorized by their size. A **computer network** is mainly of **three types**:

- ❑ LAN (Local Area Network)
- ❑ MAN (Metropolitan Area Network)
- ❑ WAN (Wide Area Network)

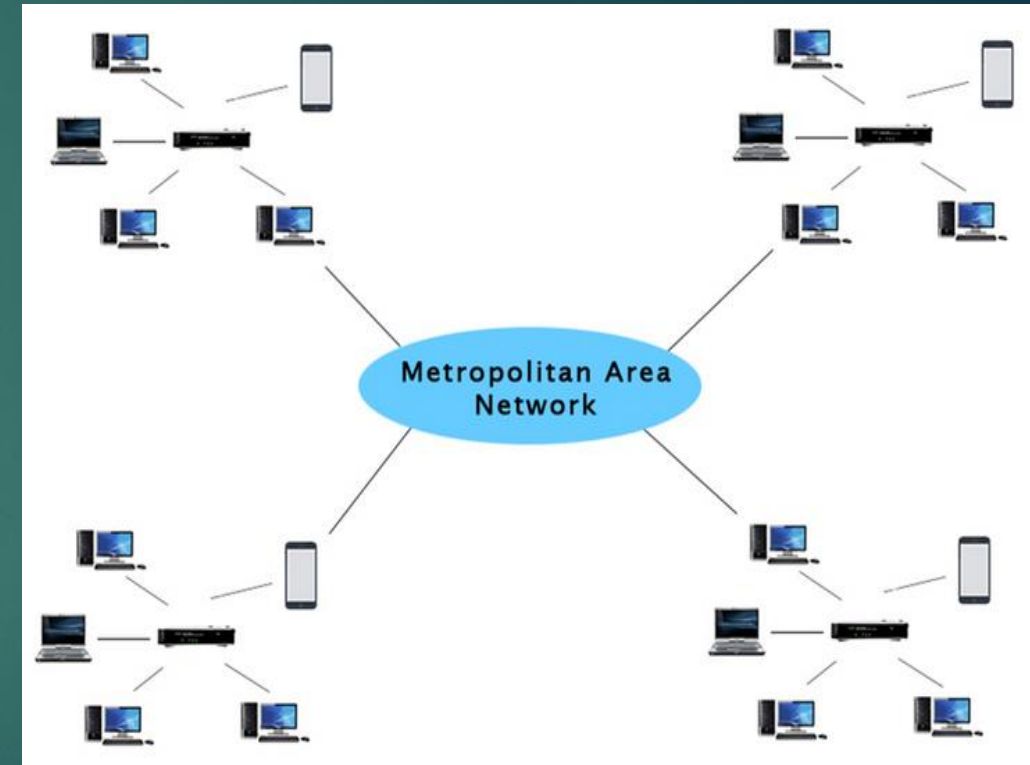
LAN (Local Area Network)

- ❑ Local Area Network is a group of computers connected to each other in a small area such as building, office.
- ❑ The data is transferred at an extremely faster rate in Local Area Network.
- ❑ Local Area Network provides higher security.



MAN (Metropolitan Area Network)

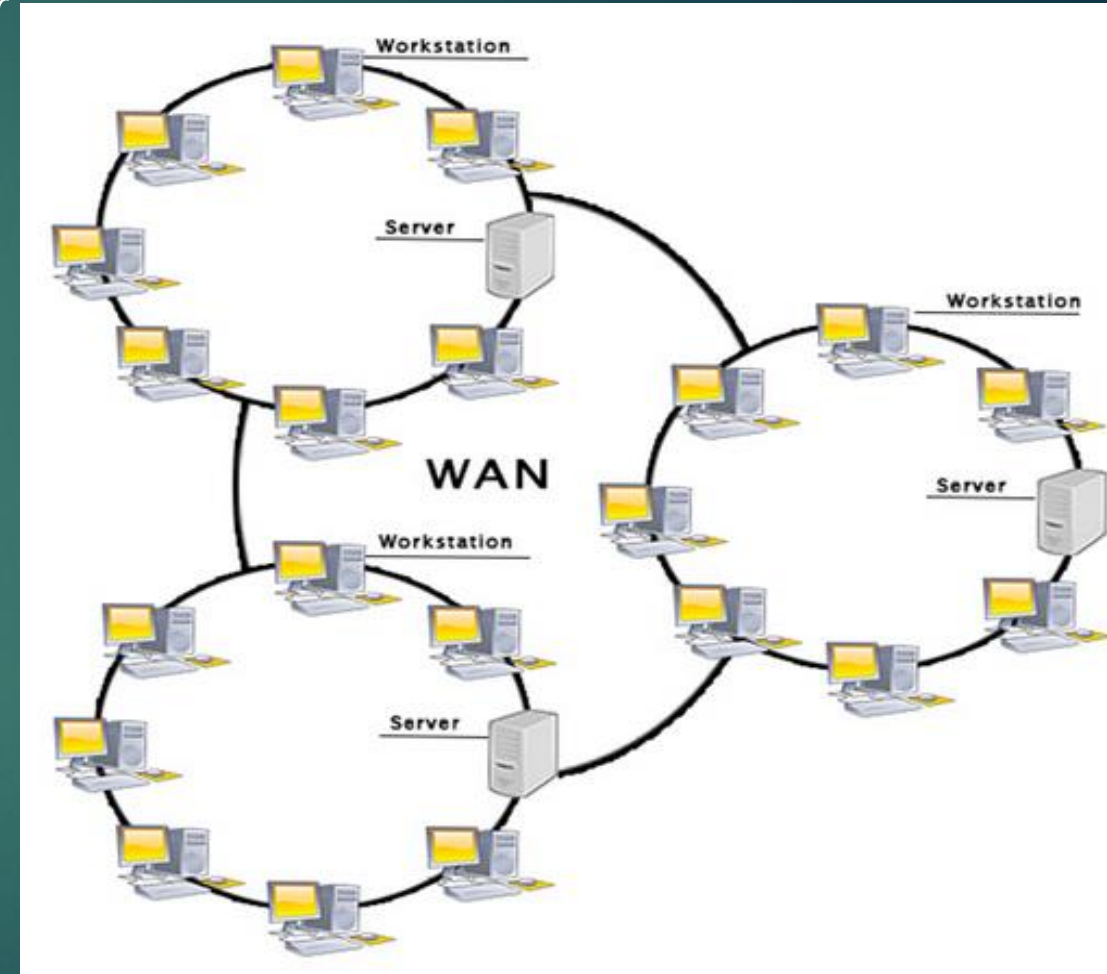
- ❑ A MAN is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- ❑ In MAN, various LANs are connected to each other.
- ❑ It has a higher range than LAN.



WAN(Wide Area Network)

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- ❑ A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- ❑ A WAN is quite bigger network than the LAN.
- ❑ A WAN is not limited to a single location, but it spans over a large geographical area through a telephone line, fiber optic cable or satellite links.
- ❑ The internet is one of the biggest WAN in the world.
- ❑ A WAN is widely used in the field of Business, government, and education.



Network Protocols

❑ A network protocol is an established set of rules that determine how:

- ❑ data is transmitted between different devices in the network.
- ❑ devices communicate with each other.
- ❑ send data between different devices .
- ❑ deal with errors in the network.

Types of Protocols

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❑ **Transmission Control Protocol (TCP):** TCP is a popular communication protocol which is used for communicating over a network. It divides any message into series of packets that are sent from source to destination and there it gets reassembled at the destination.

❑ **Internet Protocol (IP):** IP is designed explicitly as addressing protocol. It is mostly used with TCP. The IP addresses in packets help in routing them through different nodes in a network until it reaches the destination system

Types of Protocols

- ❑ **File Transfer Protocol (FTP)** :File Transfer Protocol enables file sharing between hosts. The connection is used to transfer the actual file.
- ❑ **Hyper Text Transfer Protocol (HTTP)**: HTTP is designed for transferring a hypertext. World Wide Web used HTTP as a request and response type protocol, the client sends a request to the server, which is then processed by the server before sending a response back to the client.

World Wide Web (www)



World Wide Web

- ❑ World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet.
- ❑ Websites contain text pages, digital images, audios, videos, etc.
- ❑ Users can access the content of these sites from any part of the world over the internet using their devices.
- ❑ The building blocks of the Web are web pages which are formatted in HTML and connected by links called "hypertext" or hyperlinks and accessed by HTTP.



Uniform Resource Locator (URL)

- ❑ URL is web page address.
- ❑ A particular collection of web pages that belong to a specific URL is called a website
 - ❑ www.facebook.com, www.google.com
- ❑ Small websites store all of their WebPages on a single server, but big websites or organizations place their WebPages on different servers in different countries so that when users of a country search their site they could get the information quickly from the nearest server.



Difference between World Wide Web and Internet

- ❑ Internet It is a worldwide network of devices like computers, laptops, tablets, etc. It enables users to send emails to other users and chat with them online. For example, when you send an email or chatting with someone online, you are using the internet.
- ❑ But, World Wide Web it is a webpage request from your computer using a browser, and the server Sends that page to your browser.

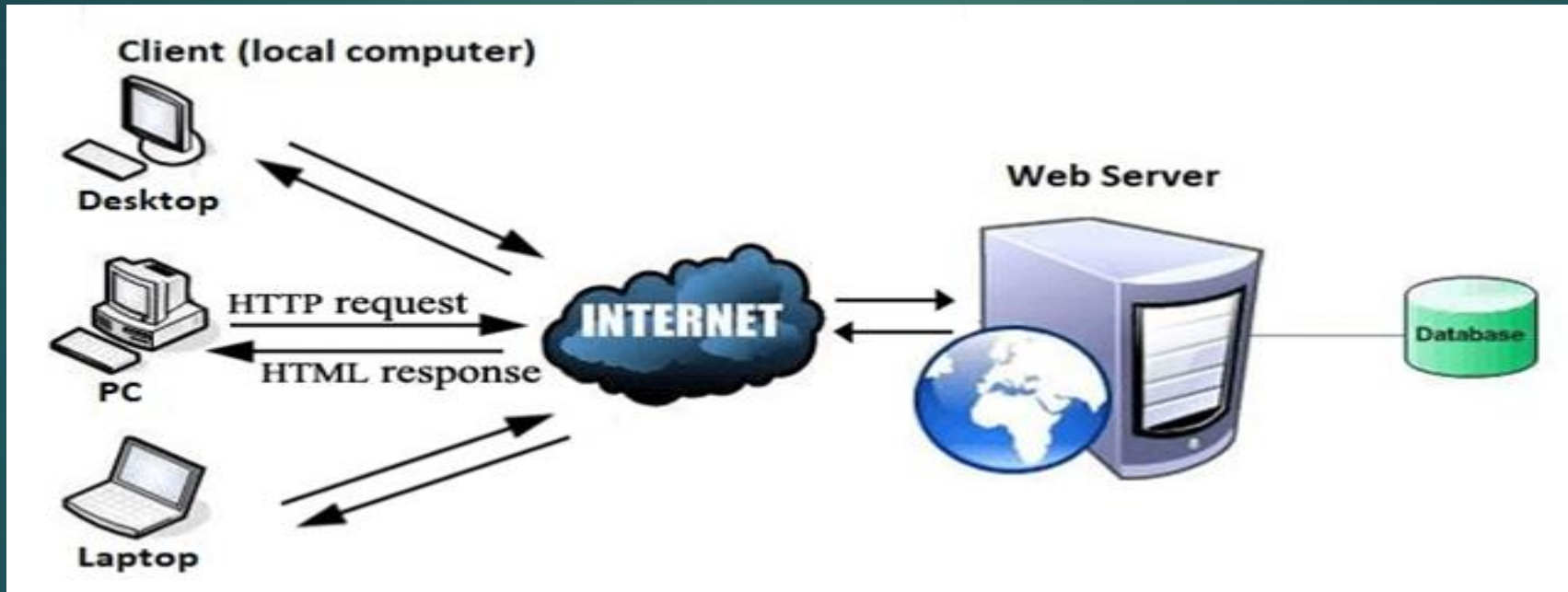


How the World Wide Web Works

- ❑ The Web works as per the internet's basic client-server format.
- ❑ The servers store and transfer web pages or information to user's computers on the network when requested by the users.
- ❑ A web server is a software program which serves the web pages requested by web users using a browser.
- ❑ The computer of a user who requests documents from a server is known as a client.
- ❑ Browser, which is installed on the user' computer, allows users to view the retrieved documents.



How the World Wide Web Works



Technologies of Transferring Web Pages

- ❑ The moment you open the browser and type a URL in the address bar or search something on Google, the WWW starts working.
- ❑ There are three main technologies involved in transferring (web pages) from servers to clients .
 - ❑ Hypertext Markup Language (HTML)
 - ❑ Hypertext Transfer Protocol (HTTP)
 - ❑ Web browsers.



Cloud Computing

Lecture Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Define cloud computing and explain its key characteristics.
- ▶ Identify cloud service models (IaaS, PaaS, SaaS).
- ▶ Explain cloud deployment models (Public, Private, Hybrid, Community).
- ▶ Discuss benefits and challenges of cloud computing.
- ▶ Describe real-world applications of cloud services.

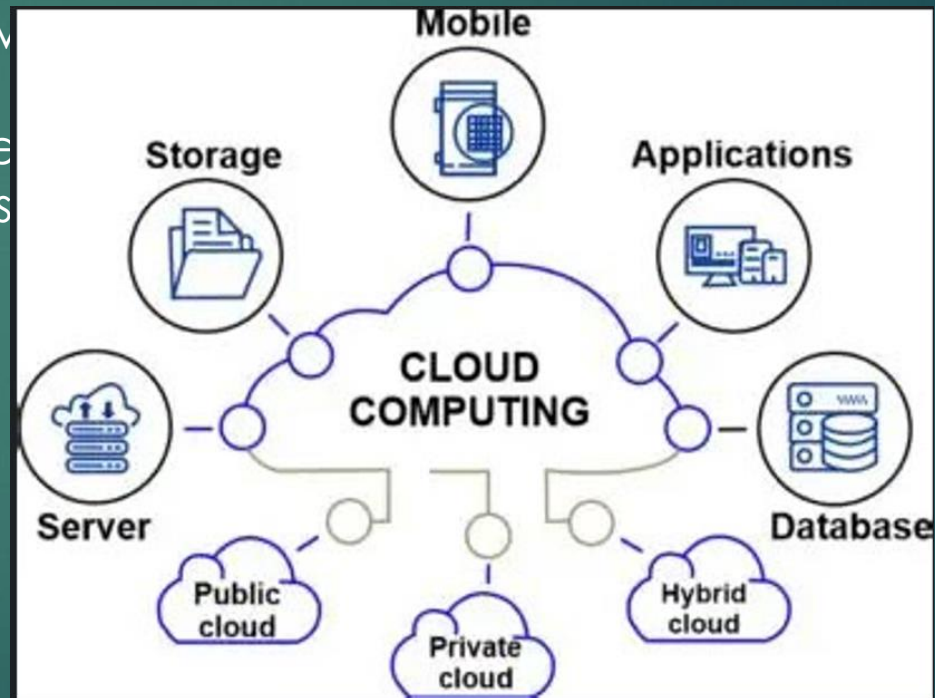
What Is Cloud Computing?

- ▶ **Definition:**

Cloud computing is the on-demand delivery of computing services—such as servers, storage, databases, networking, software over the internet on a pay-as-you-go basis.

- ▶ **Key idea:**

You use IT resources **without owning or managing the hardware.**



Key Characteristics of Cloud Computing

- ▶ On-demand self-service
- ▶ Broad network access
- ▶ Resource pooling
- ▶ Rapid elasticity
- ▶ Measured service (pay-as-you-go)
- ▶ Scalability

Cloud Service Models

- ▶ Infrastructure as a Service (IaaS)
- ▶ **Platform as a Service (PaaS)**
- ▶ Software as a Service (SaaS)

Infrastructure as a Service (IaaS)

- ▶ Provides virtual machines, storage, networks
- ▶ Example: Amazon EC2, Google Compute Engine

Platform as a Service (PaaS)

- ▶ Provides a platform for developing applications
- ▶ Example: Google App Engine, Microsoft Azure App Service

Software as a Service (SaaS)

- ▶ Software delivered over the web
- ▶ Example: Gmail, Microsoft 365, Salesforce

Cloud Deployment Models

1. Public Cloud

- ▶ Resources provided by third-party providers
- ▶ Example: AWS, Azure, Google Cloud

Cloud Deployment Models



2. Private Cloud

- ▶ Exclusive use by a single organization
- ▶ Higher security and control

Cloud Deployment Models

- ▶ **3. Hybrid Cloud**
- ▶ Combination of public + private
- ▶ Move workloads between clouds as needed

Cloud Deployment Models

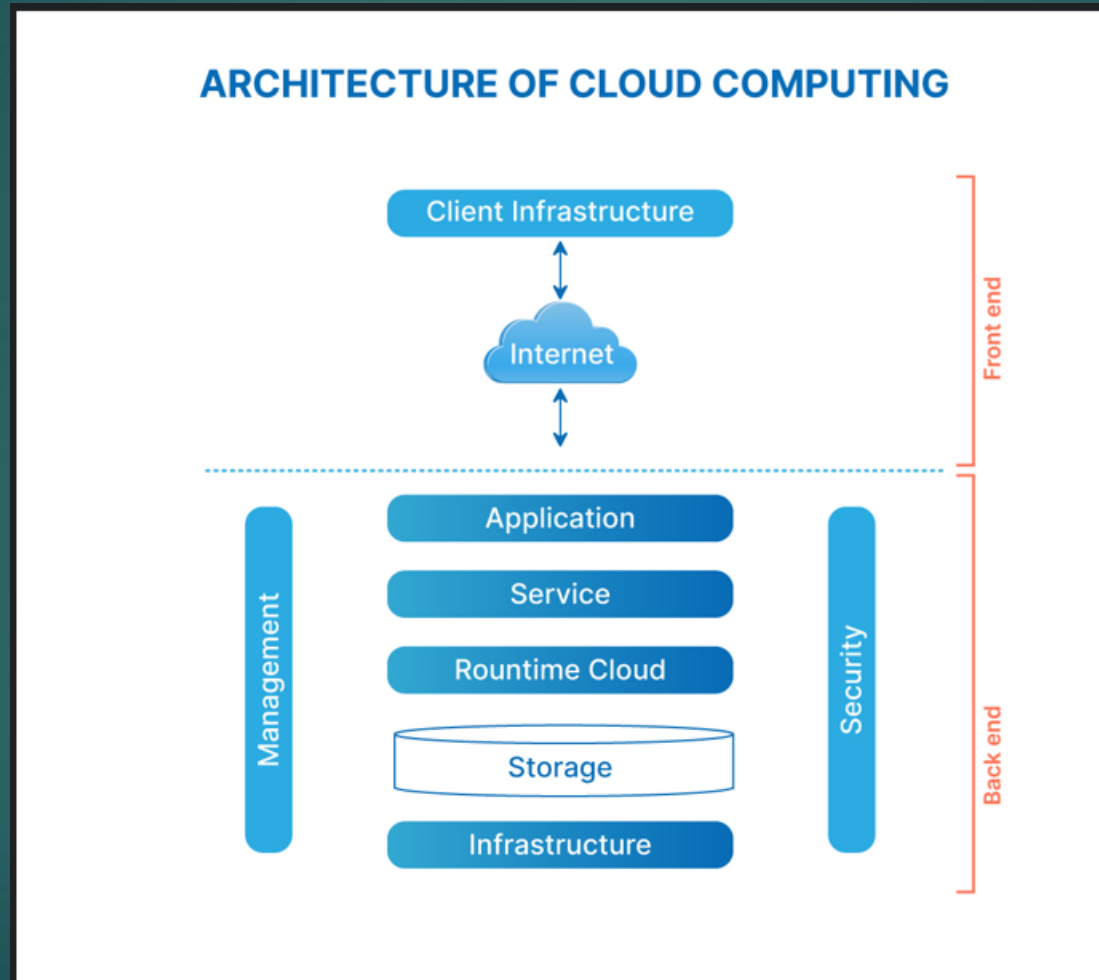
4. Community Cloud

- ▶ Shared by organizations with common needs (e.g., hospitals, banks)

Cloud Computing Architecture

- ▶ The cloud architecture is divided into 2 parts
 - ▶ Frontend: refers to the client side of cloud computing system. Means it contains all the user interfaces and applications which are used by the client to access the cloud computing services/resources. For example, use of a web browser to access the cloud platform.
 - ▶ Backend: refers to the cloud itself which is used by the service provider. It contains the resources as well as manages the resources and provides security mechanisms. Along with this, it includes huge storage, virtual applications, virtual machines, traffic control mechanisms, deployment models, etc.

Cloud Computing Architecture



Components of Cloud Computing Architecture

- ▶ **Client Infrastructure:** Client Infrastructure is a part of the frontend component. It contains the applications and user interfaces which are required to access the cloud platform. In other words, it provides a GUI(Graphical User Interface) to interact with the cloud.
- ▶ **Application :** Application is a part of backend component that refers to a software or platform to which client accesses. Means it provides the service in backend as per the client requirement.
- ▶ **Service:** Service in backend refers to the major three types of cloud based services like [SaaS](#), [PaaS](#) and [IaaS](#). Also manages which type of service the user accesses.

Components of Cloud Computing Architecture

- ▶ **Runtime Cloud:** Runtime cloud in backend provides the execution and Runtime platform/environment to the Virtual machine.
- ▶ **Storage:** Storage in backend provides flexible and scalable storage service and management of stored data.
- ▶ **Infrastructure:** Cloud Infrastructure in backend refers to the hardware and software components of cloud like it includes servers, storage, network devices, virtualization software etc.

Components of Cloud Computing Architecture

- ▶ **Management:** Management in backend refers to management of backend components like application, service, runtime cloud, storage, infrastructure, and other security mechanisms etc.
- ▶ **Security:** Security in backend refers to implementation of different security mechanisms in the backend for secure cloud resources, systems, files, and infrastructure to end-users.
- ▶ **Internet:** Internet connection acts as the medium or a bridge between frontend and backend and establishes the interaction and communication between frontend and backend.

Components of Cloud Computing Architecture

- ▶ **Database:** Database in backend refers to provide database for storing structured data, such as SQL and NOSQL databases. Example of Databases services include Amazon RDS, Microsoft Azure SQL database and Google Cloud SQL.
- ▶ **Networking:** Networking in backend services that provide networking infrastructure for application in the cloud, such as load balancing, DNS and virtual private networks.
- ▶ **Analytics:** Analytics in backend service that provides analytics capabilities for data in the cloud, such as warehousing, business intelligence and machine learning.

Benefits of Cloud Computing

- ▶ Cost savings (no hardware or maintenance)
- ▶ Scalability and elasticity
- ▶ Global access
- ▶ High availability and reliability
- ▶ Automatic updates
- ▶ Faster deployment

Challenges of Cloud Computing

- ▶ Security and privacy concerns
- ▶ Internet dependency
- ▶ Data loss and downtime risks
- ▶ Vendor lock-in

Cloud Providers

- ▶ Major cloud service providers:
 - ▶ **Amazon Web Services (AWS)**
 - ▶ **Microsoft Azure**
 - ▶ **Google Cloud Platform (GCP)**
 - ▶ **IBM Cloud**
 - ▶ **Oracle Cloud**

Cloud Applications in Business

- ▶ Examples:
- ▶ Online storage (Google Drive, OneDrive)
- ▶ Customer Relationship Systems (Salesforce)
- ▶ ERP systems
- ▶ E-commerce platforms
- ▶ Online collaboration tools (Zoom, Teams)

Summary

- ▶ Cloud = computing resources delivered online
- ▶ Service models: **IaaS, PaaS, SaaS**
- ▶ Deployment models: **Public, Private, Hybrid, Community**
- ▶ Cloud enables cost savings, flexibility, and scalability
- ▶ Challenges include security and reliability

Discussion Questions

- ▶ Why are companies moving to the cloud?
- ▶ What risks should organizations consider before adopting cloud services?
- ▶ Which cloud model (IaaS, PaaS, SaaS) do you use most in your daily life?